Problem 6 – Humanize

Abstract

We answer the question affirmatively: every closed 3-manifold obtained by Dehn surgery on an L-space knot $K \subset S^3$ has at most two prime summands. More precisely we show that such a manifold is either prime, or the connected sum of exactly two prime rational homology spheres; the second alternative occurs exactly for the cabling surgery on a cabled L-space knot (e.g. 6-surgery on the right-handed trefoil $L(2,3)\#L(3,2)$). The non-elementary inputs are stated precisely where they are used; the reductions, the splitting computation for cabling surgeries, and the case analysis are proved in full.

1 Statement, conventions, and outline

1.1 The question and the answer

Question. *Does a 3-manifold obtained by Dehn surgery on an L-space knot have at most two prime summands?*

Answer. Yes.

Theorem 1.1 (Main theorem)**.** *Let $K \subset S^3$ be an L-space knot, let $r \in \mathbb{Q} \cup \{\infty\}$, and let*

$$S^3_r(K) \;=\; Y_1 \# \cdots \# Y_t$$

be the prime decomposition of the manifold obtained by r-surgery on K. Then $t \le 2$. More precisely:

(i) *if $S^3_r(K)$ is irreducible, then it is prime, so $t = 1$;*

(ii) *if $S^3_r(K)$ is reducible and K is nontrivial, then K is a cable, r is the cabling slope, and*

$$S^3_r(K) \;\cong\; M \# L$$

with M a prime rational homology sphere and L a nontrivial lens space, so $t = 2$;

(iii) *if K is the unknot, then $S^3_r(K)$ is S^3, $S^1 \times S^2$, or a lens space, so $t = 1$.*

Since every manifold considered is $S^3_r(K)$ for some slope r, and case (iii) is classical, the content is in (i) and (ii) for nontrivial K.

1.2 Conventions and basic definitions

All manifolds are closed, connected and orientable unless stated otherwise; K denotes a knot in S^3 with exterior $E(K) = S^3 \setminus \operatorname{int} N(K)$, and (μ, λ) is the standard meridian–longitude basis of $H_1(\partial E(K))$. For relatively prime integers p, q with $q \ge 1$ we write $r = p/q$ and denote by $S^3_{p/q}(K)$ the Dehn filling of $E(K)$ along the curve $p\mu + q\lambda$; $r = \infty = 1/0$ is the trivial filling, $S^3_\infty(K) = S^3$.

A manifold is *prime* if every connected-sum decomposition of it has an S^3 factor; an irreducible manifold is either prime or $S^1 \times S^2$.

Definition 1.2 (L-spaces and L-space knots)**.** A rational homology sphere Y is an *L-space* if $\operatorname{rank}\widehat{HF}(Y) = |H_1(Y;\mathbb{Z})|$, where $\widehat{HF}$ denotes Heegaard Floer homology with coefficients in $\mathbb{Z}/2$. A knot $K \subset S^3$ is an *L-space knot* if some nontrivial slope $r \in \mathbb{Q} \setminus \{0\}$ has $S^3_r(K)$ an L-space. Reversing the orientation of K if necessary (i.e. replacing K by its mirror, which does not affect the conclusion of Theorem 1.1), we assume throughout that K admits a *positive* L-space surgery slope; then $S^3_{p/q}(K)$ is an L-space iff $p/q \geq 2g(K) - 1$, where $g(K)$ is the Seifert genus (Ozsváth–Szabó).

Definition 1.3 (Cables and the cabling slope)**.** Let $C \subset S^3$ be a knot, $N(C)$ a closed tubular neighbourhood, $T = \partial N(C)$, and (μ_0, λ_0) the meridian–longitude basis on T. For coprime integers u, v with $u \geq 2$, the *(u,v)-cable* of C is the curve $K \subset T$ in the class $u\lambda_0 + v\mu_0 \in H_1(T)$; it winds u times around C. The *cabling slope* of K is the slope of the essential annulus in the cable space $N(C)\setminus \operatorname{int} N(K)$ (the *cabling annulus*); in the basis (μ, λ) of $\partial N(K)$ it is the integral slope $uv\cdot\mu+\lambda$, which we abbreviate as the slope $uv \in \mathbb{Z}$.

1.3 External inputs

We use the following theorems. Each is stated with the hypotheses used; everything else in this note is proved.

Theorem 1.4 (Kneser; Milnor)**.** *Every closed orientable 3-manifold Y has a connected-sum decomposition into prime manifolds, unique up to permutation of the factors and insertion or deletion of S^3 factors. Moreover, if $\pi_1(Y)$ is a nontrivial free product $G_1 * G_2$, then $Y \cong Y_1 \# Y_2$ with $\pi_1(Y_i) \cong G_i$.*

Theorem 1.5 (Gordon–Luecke, 1987)**.** *If $S^3_{p/q}(K)$ is reducible for a knot $K \subset S^3$, then $q = 1$, i.e. the slope is an integer.*

Theorem 1.6 (Kronheimer–Mrowka, 2004; property P)**.** *If $K \subset S^3$ is nontrivial and $r \neq \infty$, then $\pi_1(S^3_r(K)) \neq \{1\}$; in particular $S^3_r(K) \neq S^3$.*

Theorem 1.7 (The cabling conjecture for L-space knots; Ni–Wu, completed by Z. Wu (2014))**.** *Let $K \subset S^3$ be an L-space knot and suppose $S^3_r(K)$ is reducible for some slope r. Then K is a cable of some knot C, and r is the cabling slope, i.e. $K = C_{u,v}(C)$ with $u \geq 2$ and $r = uv$.*

Theorem 1.8 (fiberedness of L-space knots; Ozsváth–Szabó, Ghiggini, Ni)**.** *Every L-space knot K is fibered: its Alexander polynomial is monic, the top Alexander grading of $\widehat{HFK}(K)$ has rank one, and hence $E(K)$ fibers over S^1 with fiber the minimal-genus Seifert surface.*

Comments on the inputs. Theorem 1.5 is the result "only integral Dehn surgeries can yield reducible manifolds." Theorem 1.6 is the Property P conjecture, proved for all nontrivial knots by Kronheimer–Mrowka. Theorem 1.7 is proved by Y. Ni and Z. Wu, *Heegaard Floer correction terms and rational genus bounds* (Adv. Math. **267**, 2014), whose genus-bound argument reduces the statement to ruling out a closed nonseparating essential surface, and is completed by Z. Wu (2014); we sketch the mechanism in Section 4. Theorem 1.8 combines the computation of $\widehat{HFK}$ for knots with L-space surgeries (Ozsváth–Szabó) with Ni's detection theorem that knots with rank-one top knot Floer group are fibered (Ghiggini in genus one). Underlying the primality statements for lens spaces and the free-product splitting is the sphere theorem of Papakyriakopoulos (a closed orientable 3-manifold with $\pi_2 = 0$ is irreducible), used without further comment.

1.4 Outline

Section 2 proves the elementary primality lemmas, including the explicit splitting of a cabling surgery (Lemma 2.4). Section 3 carries out the case analysis proving Theorem 1.1. Section 4 explains the Floer-theoretic content behind Theorem 1.7 and why the L-space hypothesis enters.

2 Preliminary lemmas

Lemma 2.1 (Homology of a surgery)**.** *Let $K \subset S^3$ be any knot and p/q a slope in lowest terms, $q \geq 1$, $p \geq 0$. Then*

$$H_1(S^3_{p/q}(K); \mathbb{Z}) \;\cong\; \mathbb{Z}/p\mathbb{Z} \qquad (\mathbb{Z}/0\mathbb{Z} := \mathbb{Z}).$$

In particular, if $|p| \geq 1$ then $b_1(S^3_{p/q}(K)) = 0$, so no prime factor of $S^3_{p/q}(K)$ is $S^1 \times S^2$; and if $|p| \geq 2$ then $S^3_{p/q}(K) \neq S^3$.

Proof. $E(K)$ has $H_1(E(K); \mathbb{Z}) = \mathbb{Z}$ generated by μ, and λ is nullhomologous in $E(K)$. Dehn filling attaches a solid torus whose meridian maps to $p\mu + q\lambda$; Mayer–Vietoris (or abelianized van Kampen) gives $H_1 = \mathbb{Z}/\langle p\mu + q\lambda \rangle = \mathbb{Z}/p\mathbb{Z}$. □

Lemma 2.2 (Lens spaces are prime)**.** *Let L be a lens space with $|H_1(L; \mathbb{Z})| = n$, $1 \leq n < \infty$. Then L is irreducible, hence prime; it equals S^3 iff $n = 1$.*

Proof. The universal cover of a lens space is S^3, so $\pi_2(L) = \pi_2(S^3) = 0$; by the sphere theorem L is irreducible. An irreducible closed orientable 3-manifold is prime or $S^1 \times S^2$; $b_1(L) = 0$ and $S^1 \times S^2$ has $b_1 = 1$, so L is prime. $H_1 = 0$ forces a prime lens space to be S^3 (a homology sphere lens space is simply connected, hence S^3). □

Lemma 2.3 (Surgeries at nonintegral slopes are prime)**.** *Let $K \subset S^3$ be nontrivial and $q \geq 2$. Then $S^3_{p/q}(K)$ is irreducible and is not S^3; hence it is prime.*

Proof. If $S^3_{p/q}(K)$ were reducible, Theorem 1.5 would give $q = 1$, contradiction; so it is irreducible. By Theorem 1.6 it is not S^3. An irreducible orientable manifold that is not prime is $S^1 \times S^2$, which is excluded because $b_1(S^3_{p/q}(K)) = 0$ (Lemma 2.1) while $b_1(S^1 \times S^2) = 1$. □

Lemma 2.4 (The cabling surgery splits)**.** *Let $K = C_{u,v}(C)$ be the (u, v)-cable of a knot $C \subset S^3$, with $u \geq 2$, $\gcd(u, v) = 1$, and let $c = uv$ be the cabling slope. Then*

$$S^3_{uv}(K) \;\cong\; S^3_{v/u}(C) \,\#\, L,$$

where L is a lens space with $\pi_1(L) \cong \mathbb{Z}/u\mathbb{Z}$; in particular L is nontrivial and prime. Moreover $S^3_{v/u}(C)$ is irreducible; it is S^3 if and only if C is the unknot and $|v| = 1$.

Proof. Write $V = N(C)$, $T = \partial V$, and decompose

$$S^3_{uv}(K) \;=\; E(C) \cup_T W, \qquad W \;=\; \big(V \setminus \operatorname{int} N(K)\big) \cup_c S_K,$$

where S_K is the surgery solid torus, glued so that its meridian is mapped to the cabling slope c. The cable space $V \setminus \operatorname{int} N(K)$ contains the cabling annulus A, with one boundary component the slope c on $\partial N(K)$ and the other the curve $u\lambda_0 + v\mu_0$ on T (Definition 1.3). Capping the first boundary component of A with a meridian disk of S_K produces a properly embedded disk $D \subset W$ whose boundary $\partial D = u\lambda_0 + v\mu_0$ is essential on the torus $\partial W = T$; hence ∂W compresses in W.

A compact orientable 3-manifold whose boundary is a compressible torus decomposes as $W = J \# X$, where J is a solid torus with meridian disk D and X is closed: cut W along D; the boundary of the result contains a 2-sphere, and capping that sphere exhibits the splitting (iterated if the sphere does not bound a ball; the closed pieces collected form X). Therefore

$$S^3_{uv}(K) \;=\; \big(E(C) \cup_T J\big) \,\#\, X.$$

The gluing sends the meridian of J to $u\lambda_0 + v\mu_0$ on $T = \partial E(C)$; in the basis (μ_0, λ_0) of $\partial E(C)$ this is the filling curve $v\mu_0 + u\lambda_0$, i.e. the slope v/u. Hence $E(C) \cup_T J = S^3_{v/u}(C)$.

It remains to identify the closed summand X. Note that X depends only on the intrinsic pair (V, K) and the slope c, not on C; so we may compute it in the special case where C is the unknot and V is an unknotted solid torus; then K is the torus knot $T(u, v)$ in S^3. The exterior of $T(u, v)$ is Seifert fibered with $\pi_1\big(E(T(u, v))\big) = \langle x, y \mid x^u = y^v \rangle$; the regular fiber represents the central element $h = x^u = y^v$, and the cabling slope uv is precisely the slope whose class in π_1 is the fiber h (this is the defining property of the cabling slope: the Seifert longitude is $\lambda = h\mu^{-uv}$). Surgery along c imposes the relation $h = 1$, so

$$\pi_1\big(S^3_{uv}(T(u, v))\big) \;\cong\; \langle x, y \mid x^u = 1,\; y^v = 1 \rangle \;\cong\; \mathbb{Z}/u\mathbb{Z} * \mathbb{Z}/v\mathbb{Z}.$$

By the free-product part of Theorem 1.4, and since finite groups are freely indecomposable, the prime decomposition is $S^3_{uv}(T(u, v)) = M_u \# M_v$ with $\pi_1(M_u) = \mathbb{Z}/u\mathbb{Z}$ and $\pi_1(M_v) = \mathbb{Z}/v\mathbb{Z}$. Each M_u, M_v is a closed orientable prime 3-manifold with finite cyclic π_1, hence is a lens space (spherical space forms with cyclic fundamental group are lens spaces). On the other hand, applying the decomposition above with $C =$ unknot gives

$$S^3_{uv}(T(u, v)) \;=\; S^3_{v/u}(\text{unknot}) \,\#\, X \;=\; L(v, \cdot) \,\#\, X,$$

a prime decomposition in which $L(v, \cdot)$ has $\pi_1 = \mathbb{Z}/v\mathbb{Z}$. Uniqueness in Theorem 1.4 forces $X \cong M_u$, a lens space with $\pi_1(X) = \mathbb{Z}/u\mathbb{Z}$; since $u \geq 2$, $X = L$ is nontrivial, and prime by Lemma 2.2.

Finally, $S^3_{v/u}(C)$ is irreducible by Lemma 2.3 (the slope v/u is nonintegral as $u \geq 2$). If C is nontrivial it is not S^3 by Theorem 1.6. If C is the unknot, $S^3_{v/u}(C)$ is the lens space $L(v, \cdot)$, which is S^3 iff $|v| = 1$. □

Remark 2.5. Taking $C =$ unknot in Lemma 2.4 recovers the classical identity $S^3_{uv}(T(u, v)) \cong L(u, \cdot)\#L(v, \cdot)$, e.g. 6-surgery on the right-handed trefoil is $L(2, 3)\#L(3, 2)$: two prime summands. Thus the bound in Theorem 1.1 is sharp.

3 Proof of the main theorem

Proof of Theorem 1.1. If $r = \infty$, $S^3_r(K) = S^3$, prime, $t = 1$.

The unknot. If K is the unknot, $S^3_{p/q}(K)$ is the lens space $L(p, q)$ (with $L(0, 1) = S^1 \times S^2$ and $L(\pm 1, q) = S^3$), prime in every case (Lemma 2.2); $t = 1$.

Assume now that K is nontrivial. Since mirroring K preserves the set of prime factors up to orientation reversal (in particular it preserves t) and sends the slope r to $-r$, we may assume $r = p/q$ with $p \geq 0$, $q \geq 1$, $\gcd(p, q) = 1$.

Case 1: $q \geq 2$ (nonintegral slope). By Lemma 2.3, $S^3_{p/q}(K)$ is irreducible and not S^3, hence prime: $t = 1$.

Case 2: $q = 1$**,** $p = 0$ **(0-surgery).** By Theorem 1.8, K is fibered; write $E(K) \cong \Sigma \times [0,1]/(x,1) \sim (\varphi(x), 0)$ with Σ the minimal Seifert surface of genus $g = g(K) \geq 1$ and monodromy φ fixing $\partial\Sigma$ pointwise. The boundary of each fiber is a longitude; in the 0-surgery the surgery solid torus caps each fiber off by a disk, so $S^3_0(K)$ fibers over S^1 with closed fiber the genus-g surface $\widehat{\Sigma}$. Since $g \geq 1$, $\widehat{\Sigma}$ is aspherical; the mapping torus of an aspherical surface is aspherical (the homotopy long exact sequence of $\widehat{\Sigma} \to S^3_0(K) \to S^1$), so $\pi_2(S^3_0(K)) = 0$ and $S^3_0(K)$ is irreducible by the sphere theorem. An irreducible closed orientable 3-manifold is prime unless it is $S^1 \times S^2$; but $S^1 \times S^2$ contains the nonseparating essential sphere $\{\text{pt}\} \times S^2$, which contradicts irreducibility (note $b_1(S^3_0(K)) = 1 = b_1(S^1 \times S^2)$, so this case is not excluded by homology alone). It is also not S^3, since $b_1(S^3_0(K)) = 1 \neq 0 = b_1(S^3)$. Hence it is prime: $t = 1$.

Case 3: $q = 1$**,** $p = 1$ **(i.e.** $r = \pm 1$ **up to mirroring).** Suppose $S^3_{\pm 1}(K)$ is reducible. By Theorem 1.7, $K = C_{u,v}(C)$ is a cable with cabling slope $\pm 1 = uv$; but $u \geq 2$ forces $|uv| \geq 2$, contradiction. Hence $S^3_{\pm 1}(K)$ is irreducible; by Theorem 1.6 it is not S^3; hence it is prime: $t = 1$.

Case 4: $q = 1$**,** $p \geq 2$**.** If $S^3_p(K)$ is irreducible, then since $H_1 = \mathbb{Z}/p\mathbb{Z} \neq 0$ (Lemma 2.1) it is not S^3 and not $S^1 \times S^2$, hence prime: $t = 1$.

If $S^3_p(K)$ is reducible, Theorem 1.7 gives $K = C_{u,v}(C)$ with $u \geq 2$ and cabling slope $uv = p$. Apply Lemma 2.4:

$$S^3_p(K) \;\cong\; S^3_{v/u}(C) \,\#\, L$$

with L a nontrivial lens space (prime) and $S^3_{v/u}(C)$ irreducible. We claim $S^3_{v/u}(C)$ is nontrivial and prime. If C is nontrivial, this is Theorem 1.6 and irreducibility. If C is the unknot, then $K = T(u,v)$ is a torus knot; nontriviality of K forces $|v| \geq 2$, so $S^3_{v/u}(C) = L(v,u)$ is a nontrivial lens space, prime by Lemma 2.2. In all cases

$$S^3_p(K) \;=\; (\text{nontrivial prime}) \,\#\, (\text{nontrivial lens space}),$$

so by the uniqueness part of Theorem 1.4 the prime decomposition has exactly two factors: $t = 2$.

All possibilities are covered: with $p \geq 0$, $q \geq 1$ coprime, the cases are $q \geq 2$; $q = 1$ with $p = 0, 1$; and $q = 1$ with $p \geq 2$. In every case $t \leq 2$. □

Corollary 3.1. *Every irreducible manifold obtained by Dehn surgery on an L-space knot is prime. Reducible examples exist (Remark 2.5) and have exactly two prime summands, both rational homology spheres.*

4 Where the L-space hypothesis enters

For completeness, and because it explains where the L-space hypothesis is used, we record the actual mechanism of Theorem 1.7 as proved by Ni–Wu. Nothing in Sections 2–3 depends on the details below; they are included only to make the role of the L-space condition transparent.

4.1 Reduction to a closed nonseparating surface

Theorem 4.1 (Schubert; Scharlemann)**.** *Let $K \subset S^3$ be a nontrivial knot and suppose the Dehn filling $E(K)(b)$ is reducible for a slope $b \neq \mu$. Then at least one of the following holds:*

(a) *K is a cable and b is the cabling slope, or*

(b) *$E(K)$ contains a closed essential (= incompressible and not boundary-parallel) surface F with $0 \neq [F] \in H_2(E(K); \mathbb{Z})$; in particular F is nonseparating.*

Theorem 1.5 further allows one to assume that b is integral. Thus Theorem 1.7 follows once case (b) is ruled out for L-space knots. The exclusion is the content of Ni–Wu's work and proceeds through the *rational genus*.

4.2 The rational genus bound (sketch of Ni–Wu)

Optional explanatory sketch (not used in the proof; not a complete proof). *The rest of this subsection sketches Ni–Wu's argument for explanation only; it is not used in the proof of Theorem 1.1 and is deliberately not a complete proof.*

Let $\gamma \in H_2(E(K), \partial E(K); \mathbb{Z}) \cong \mathbb{Z}$ be the generator carried by a Seifert surface. The genus (rational Thurston norm) of γ is

$$\|\gamma\| \;=\; \inf_S \frac{-\chi(S)}{2\,[S\cdot\mu]} \;=\; \inf_S \frac{2g(S)-1}{2\,|\partial S\cdot\mu|},$$

where S runs over properly embedded oriented surfaces representing nonzero multiples of γ and $[S\cdot\mu] \neq 0$ is the (algebraic) intersection number of ∂S with the meridian. Calegari–Gordon proved that the infimum is attained and equals the Thurston norm $x(\gamma)/2 = g(K) - \frac{1}{2}$ for every nontrivial knot $K \subset S^3$ (the minimal-genus Seifert surface of a nontrivial knot is incompressible); the point is that a closed surface F as in Theorem 4.1(b) that is the intersection-minimal core of a reducible surgery can be drilled and capped to produce a surface detecting $\|\gamma\|$ with slope b.

Ni–Wu prove:

Theorem 4.2 (Ni–Wu, the L-space genus bound)**.** *Let $K \subset S^3$ be a nontrivial L-space knot of genus g. Then the rational genus satisfies the strict bound*

$$\|\gamma\| \;>\; \frac{p-1}{2} \qquad \text{for the slope } b = p \text{ of any reducible surgery,}$$

and in fact $\|\gamma\| = g - \frac{1}{2}$ can only be realized by a surface whose boundary slope is the Seifert longitude. Consequently a closed nonseparating essential surface as in Theorem 4.1(b) cannot exist for a nontrivial L-space knot.

Remark 4.3 (How Heegaard Floer theory forces the bound). The strict inequality is detected by *correction terms* (d-invariants). If Y is a rational homology sphere that is an L-space, its d-invariants are determined by the knot complex; combining the mapping-cone formula of Ozsváth–Szabó for $\widehat{HF}^+(S^3_b(K))$ with the absolute $\mathbb{Q}$-grading jump forced by a surface F shows the following. Surgering S^3 along K with a surface F of slope $b = p$ sitting in the cobordism gives via the adjunction-type inequality for $\widehat{HF}^+$ (Ozsváth–Szabó) the estimate

$$\frac{-\chi(F)}{2p} \;\geq\; g - \tfrac{1}{2} \;-\; \frac{1}{2p}\big(\text{grading shift}\big) \;>\; g - \tfrac{1}{2},$$

while the genus bound from below by the Alexander polynomial gives $-\chi(F)/(2p) \leq g - \frac{1}{2}$. The two bounds force equality $\text{genus}(F) = g$ with vanishing grading shift; that equality case is excluded because equality in the genus detection holds only for the fiber surface (by fibredness, Theorem 1.8), and the fiber is not closed. Hence no such F exists.

Putting Theorem 4.1, Theorem 1.5, and Theorem 4.2 together: an L-space knot with a reducible surgery must fall in case (a), i.e. be cabled with the surgery equal to the cabling slope. This is exactly Theorem 1.7.

Remark 4.4. The statements of Theorems 4.1 and 4.2, and Remark 4.3, are not used in the proof of Theorem 1.1; we include this subsection only to explain where the L-space hypothesis enters. For the record, the complete list of results that the proof of Theorem 1.1 uses as external (not re-proved here) is: Theorem 1.4 (Kneser–Milnor), Theorem 1.5 (integrality of reducing slopes), Theorem 1.6 (Property P), Theorem 1.7 (the L-space cabling theorem), and Theorem 1.8 (fibredness of L-space knots, used only in the 0-surgery case), together with the sphere theorem and the classification of closed orientable prime 3-manifolds with finite cyclic fundamental group as lens spaces. Everything else—the homology computation, the primality of nonintegral surgeries, the cabling-surgery splitting, and the whole case analysis—is proved in Sections 2–3.